Sayantan Pal

Unmanded Vehicle System Lab, Indian Institute of Technology Kharagpur (IIT kharagpur), India

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Professional Preparation

Indian Institute Of Technology, Kharagpur	Aerospace Engineering	PhD, 2026	79.10%
IIEST, Shibpur	Mechatronics	Mtech, 2020	79.76%
MAKAUT, Kolkata	Mechanical Engineering	Btech, 2016	80.01%

Research Interests

Autonomous Vehicle Guidance	Unmanned Vehicle Control	Path Planning
Obstacle Avoidance	Missile Guidance	Machine Learning
Soft Computing	Wearable Sensors	Soft Robotics

Skills

1. **Programming & Development:** MATLAB, Python, C/C++, ROS 2
2. **Hardware & Embedded Systems:** Quanser QDrone, Quanser QBot
3. **Simulation & Motion Capture:** OptiTrack Motive
4. **CAD & Mechanical Design:** CATIA, SolidWorks
5. **Certifications:** TensorFlow for Artificial Intelligence, Machine Learning and Deep Learning | (Coursera, 2020)

Research Experience

08/2021–Present **Graduate Research Assistant**, Department of Aerospace Engineering, Indian Institute of Technology Kharagpur, India: Conducting doctoral research on the guidance and control of unmanned vehicles, with a focus on path planning, obstacle avoidance, and missile guidance.

12/2020–07/2021 **Project Associate**, Advanced Technology Development Centre (ATDC), CSIR–Central Mechanical Engineering Research Institute, India: Contributed to the design and development of a tendon-driven hand exoskeleton for post-stroke rehabilitation.

07/2019–08/2020 **Research Assistant**, Robotics and Automation Department, CSIR–Central Mechanical Engineering Research Institute, India: Conducted research on the design and development of flexible and stretchable strain sensors for a lower-limb exosuit for military applications as part of a Master's degree.

Journal Articles

[1] **S. Pal** and S. Hota, "Impact Angle Control Guidance Using Asymmetric Integral Barrier Lyapunov Function With FOV and Input Constraints," *IEEE Control Systems Letters*, vol. 9, pp. 2663–2668, 2025. Link

[2] **S. Pal** and S. Hota, "Curvature Constrained Vector Field Guidance for UAVs," *IEEE Transactions on Aerospace and Electronic Systems*, pp. 1–13, 2026. Link

[3] P. Mahajan, **S. Pal**, and S. Hota, "Continuous-Curvature Dubins-Inspired L1 Guidance with Sliding Mode Control-Based Tracking for UAVs," *Proceedings of the Institution of Mechanical Engineers, Part G: Journal of Aerospace Engineering*, 2025. Link

[4] S. Pal, N. K. Singh, and S. Hota, "Look-Angle Optimized Variable Gain Guidance Law With Impact-Angle Constraint," *International Journal of Control, Automation and Systems*. **(Under Review)**.

[5] S. Pal, N. K. Singh, and S. Hota, "Optimization-Based Composite Guidance for 3D Impact-Angle-Constrained Interception Under Seeker Field-of-View Constraints," *International Journal of Aeronautical and Space Sciences*. **(Under Review)**.

[6] A. D. Menon, S. Pal, and S. Hota, "Three-Dimensional Impact Angle Control Guidance Law with Field-of-View Constraint," *The Aeronautical Journal*. **(Under Review)**.

Peer-reviewed Conference

[1] P. P. Kumar, **S. Pal**, and S. Hota, "Barrier Sliding Mode Control for Curvature-Constrained Nonlinear Planar Path Following of UAVs," *Proceedings of the 2026 IEEE Conference on Decision and Control (CDC)*, 2026. **(Accepted)**

[2] **S. Pal** and S. Hota, "Arccosine Vector Field Guidance for Curvilinear Path Following of UAVs," *Proceedings of the International Conference on Control, Decision and Information Technologies (CoDIT)*, 2026**. (Accepted)**

[3] I. Phanse, **S. Pal**, and S. Hota, "Varying-Gain Lyapunov Guidance Vector Field for Elliptical Path Following," *Proceedings of the IEEE SPACE Conference*, 2026. **(Accepted)**

[4] P. Madure, G. Khurd, **S. Pal**, and S. Hota, "Guidance Law for Surveillance by UAVs," *Proceedings of the IEEE SPACE Conference*, 2026.**(Accepted)**

[5] K. S. K. Hrithika, **S. Pal**, and S. Hota, "UAV Path Following with Obstacle Avoidance Using Vector Field Approach," *Proceedings of INSTCon*, 2026. **(Accepted)**

[6] P. Mahajan, **S. Pal**, and S. Hota, "Time-Optimal L1 Guidance for Straight-Line Path Following in 3D," *Proceedings of the 12th International Conference on Mechatronics and Robotics Engineering (ICMRE)*, 2026. **Link**

[7] N. K. Singh, **S. Pal**, A. D. Menon, and S. Hota, "Impact Angle Constrained Guidance Laws for Stationary Target Interception in 3D Space," *Proceedings of the AIAA SCITECH 2026 Forum*, 2026. Link

[8] N. K. Sahu, **S. Pal**, and S. Hota, "Impact Angle Constrained 3D Guidance Law for Stationary Target Interception," *Proceedings of the IEEE SPACE Conference*, 2025. Link

[9] S. Singha, **S. Pal**, and S. Hota, "Smooth Trajectory Generation for Gap Traversal by UAVs," *Proceedings of the IEEE SPACE Conference*, 2025. Link

[10] P. P. Kumar, **S. Pal**, and S. Hota, "Path Convergence of UAVs via Fast Terminal Sliding Mode Control," *Proceedings of the 11th Indian Control Conference (ICC)*, 2025. Link

[11] **S. Pal** and S. Hota, "Vector Field Guidance with Curvature Constraint for UAVs," *AIAA SCITECH 2025 Forum*, 2025. Link

[12] **S. Pal** and S. Hota, "Nonlinear Control Law for Path Following by UAVs," *Proceedings of the 10th Indian Control Conference (ICC)*, pp. 476–481, 2024. Link

[13] **S. Pal** and S. Hota, "Time-Optimal Bézier Curves for Straight-Line and Circular Path Convergence in 3D Space," *Proceedings of the 9th Indian Control Conference (ICC)*, pp. 365–370, 2023. Link

[14] **S. Pal**, D. Sarkar, S. S. Roy, A. Kumar, and A. Arora, "Development of a Stretchable and Flexible Conductive Fabric-Based Sensorized Pneumatic Artificial Muscle," *Proceedings of the 4th International Conference on Electronics, Communication and Aerospace Technology (ICECA)*, pp. 339–344, 2020. Link

[15] **S. Pal**, D. Sarkar, S. S. Roy, A. Paul, and A. Arora, "Design, Development and Analysis of a Conductive Fabric-Based Flexible and Stretchable Strain Sensor," *IOP Conference Series: Materials Science and Engineering*, 2020. Link

[16] D. Sarkar, **S. Pal**, S. S. Roy, A. Kumar, and A. Arora, "Estimation of Transmission Force in Assistive Devices Using Conductive Liquid Metal-Based Sensorized Pneumatic Artificial Muscle," *Proceedings of the 5th IEEE International Conference on Recent Advances and Innovations in Engineering (ICRAIE)*, pp. 1–5, 2020. Link

[17] A. Arora, D. Sarkar, **S. Pal**, A. Kumar, S. Sen, and S. S. Roy, "A Multi-Technique Measurement and Estimation of Volume for Pneumatic Artificial Muscle," *Proceedings of the 5th IEEE International Conference on Recent Advances and Innovations in Engineering (ICRAIE)*, pp. 1–5, 2020.Link

Invited Talks

1. *"Nonlinear Guidance for Impact Angle Constrained Target Interception Under FOV and Input Constraints,"* Invited talk, **Robotics Café Seminar**, 2026.

Recognitions

1. **IEEE American Control Conference (ACC) Travel Grant** — *IEEE American Control Conference, USA* (2026).
   Awarded competitive travel and lodging funding to present research at the *IEEE American Control Conference (ACC)*.

2. **IIT Kharagpur Travel Grant** — *Indian Institute of Technology Kharagpur, India* (2026).
   Supported international paper presentation at the *IEEE International Conference on Control, Decision and Information Technologies (CoDIT), Bari, Italy*.

3. **IIT Kharagpur Travel Grant** — *Indian Institute of Technology Kharagpur, India (2025)*.
   Supported international paper presentation at the *AIAA SciTech Forum, Orlando, Florida, USA*.

4. **IEEE Indian Control Conference (ICC) Travel Grant** — *IEEE 9th Indian Control Conference, India* (2023).
   Awarded competitive travel funding to present research at the *IEEE Indian Control Conference (ICC)*.

5. **GATE Fellowship** — *Government of India* (2018–2021).
   Secured a national fellowship through the Graduate Aptitude Test in Engineering (GATE).

Professional Service & Leadership

- **Reviewer**, *IEEE Control System Letters, International Journal of Control Automation and Systems, **ICC**-2025, AIAA SciTech 2025.*
- **Graduate Member**, *IEEE*
- **Volunteer**, *Advances in Robotics and AI* 2020 CSIR-CMERI, *ICTACEM* 2026 IIT Kharagpur.